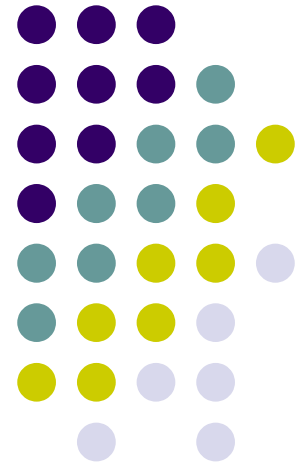


Chapter 21

Electric Fields



PHYS 2321

Week 2 on Electric Fields & Charge Distributions



Day 1 outline

- 1) Attendance
- 2) Homework for Friday
 - a) Ch. 21 read sec. 6-10(11), Ch. 22 sec. 1 (flux)
 - b) Ch. 21 Probs. 22-25, 27, 30, 33, 39, 47,48, 55,57
MisQ. 4,5,7,8,11
 - c) Try practice quizzes on web page
 - d) Watch my YouTube videos on E-fields.
- 3) Today:

Vector sums of coulomb forces (P. 21.13)

Electric Field basics, point charges.

Notes: Hwk 1 not yet graded.

****Quiz on Friday on Coulomb's law and charge.****

No office hrs today 2-3pm. Last day to add $\bar{3}$ 9/1

PHYS 2321

Week 2 on Electric Fields & Charge Distributions



Day 2 outline

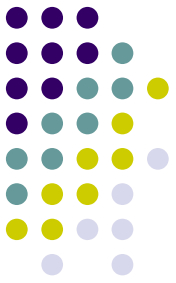
- 1) Attendance
- 2) Homework for Friday
 - a) Ch. 21 read sec. 6-10(11), Ch. 22 sec. 1 (flux)
 - b) Ch. 21 Probs. 22-25, 27, 30, 33, 39, 47,48, 55,57
MisQ. 4,5,7,8,11
 - c) Try practice quizzes on web page
 - d) Watch my YouTube videos on E-fields.
- 3) Today: Electric Field basics
 - Calculate E due to discrete (point) charges
 - Calculate E due to continuous charge distributions
(Line charge, Ring charge)

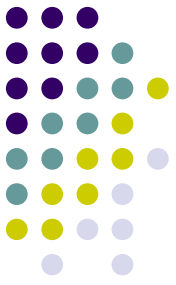
Notes: Hwk 1 mean = 9.8/10(s2), 9.7/10(s4)

****Quiz on Friday on Coulomb's law and charge.**

Electric

- The electric field
- Field for vacuum
- Faraday while studying
- Similar to
- Invisible

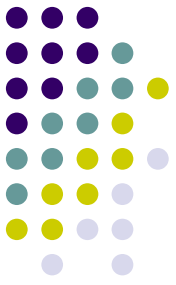




Electric Field – Definition

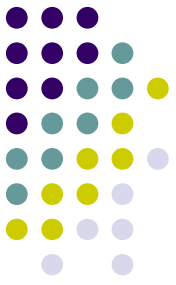
- An **electric field** is said to exist in the region of space around a charged object
 - This charged object is the **source charge**
- When another charged object, the **test charge**, enters this electric field, an electric force acts on it
- Even with NO test charge, an E-field is present and it stores energy!

Electric Field – Definition, cont



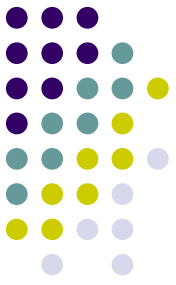
- The electric field vector, $\vec{E}$, at a point in space is defined as the electric force $\vec{F}_{\text{test}}$ acting on a positive test charge, q_{test} placed at that point, divided by the test charge:

$$\vec{E} \equiv \frac{\vec{F}_{\text{test}}}{q_{\text{test}}}$$



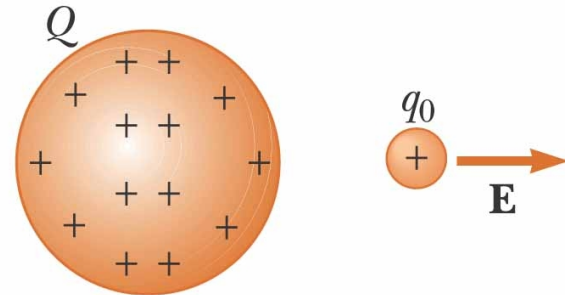
Electric Field, Notes

- $\vec{E}$ is the field produced by some charge or charge distribution called the *source charge*.
- You must not include the test charge as part of the source charge or you will get “infinity” for $\mathbf{F}_{\text{test}}$ and $\mathbf{E}$.
- The presence of the test charge is not necessary for the field to exist.
- The test charge serves as a detector of the field
 - It is small
 - It is positive
 - It is located at the point, P , of interest

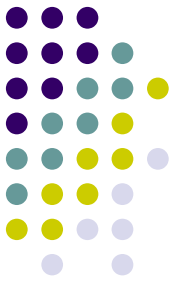


Electric Field Notes, Final

- The direction of $\vec{E}$ is that of the force on a *positive* test charge
- The SI units of $\vec{E}$ are N/C
- $\vec{E}$ points away from positive charges and towards negative
- See PhET “charges-and-fields_en”

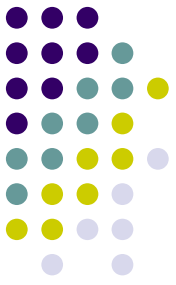


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Relationship Between $\vec{F}$ and $\vec{E}$

- $\vec{F} = q\vec{E}$
 - Valid when q is point charge, but $\vec{E}$ can be due to an extended charge or a point charge.
 - For a large test charge, the field could vary over the extent of the object
- If q is positive, the force and the field are in the same direction
- If q is negative, the force and the field are in opposite directions



Electric Field, Vector Form

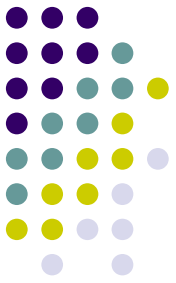
- The Coulomb force, on a test charge q_o by a point charge q_s is:

$$\vec{F}_{os} = k_e \frac{q_s q_o}{r^2} \hat{r}_{so}$$

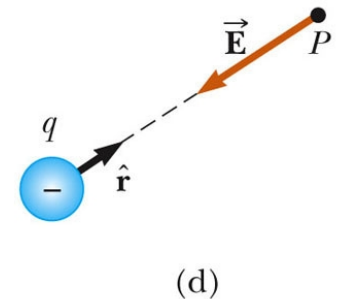
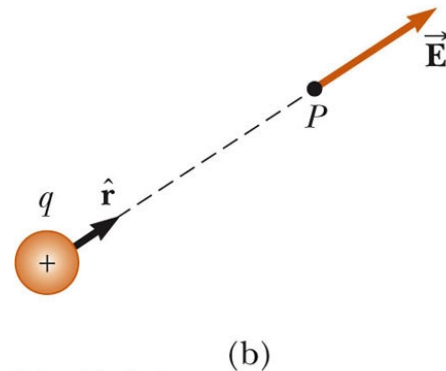
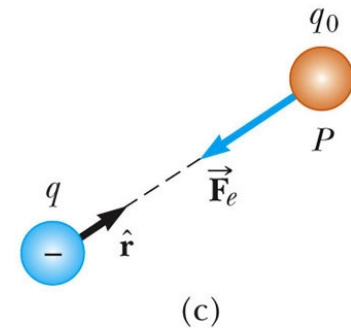
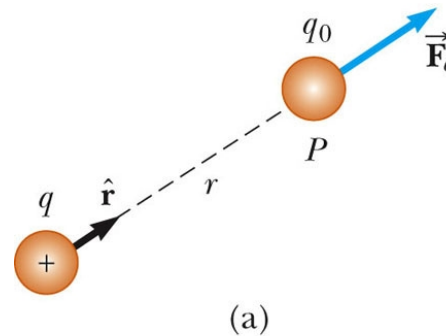
- Then, the electric field due to a point charge will be

$$\vec{E} = \frac{\vec{F}_{os}}{q_o} = k_e \frac{q_s}{r^2} \hat{r}_{so}$$

More About Electric Field Direction



- a) q is positive, the force is directed away from q
- b) The direction of the field is also away from the positive source charge
- c) q is negative, the force is directed toward q
- d) The field is also toward the negative source charge



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PHYS 2321

Week 2 on Electric Fields & Charge Distributions

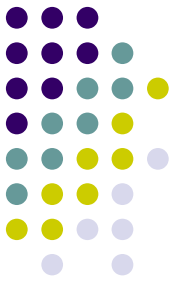


Day 3 outline

- 1) Quiz 1 (~10 minutes)
- 2) Homework for Friday
 - a) Ch. 21 read sec. 6-10(11), Ch. 22 sec. 1 (flux)
 - b) Ch. 21 Probs. 22-25, 27, 30, 33, 39, **47**, 48, 55, 57
MisQ. 4, 5, 7, 8, 11 (Due today by 2:00)
 - c) Next: Ch. 22 P. 1, 2, 5, 6, 9, 10, 13, 17, 19, 20, 35 + MisCQs
- 3) Today: Electric Fields (cont.)
 - Calculate E due to discrete (point) charges
 - Calculate E due to continuous charge distributions
(Line charges, Ring charge)

Notes:

Superposition with Electric Fields

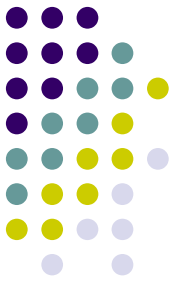


- At any point P , the total electric field due to a group of point source charges equals the vector sum of the electric fields of all the charges

$$\vec{E}_{net} = \frac{\vec{F}_{net}}{q_o} = \frac{1}{q_0} \sum_{i=1}^N \vec{F}_i$$

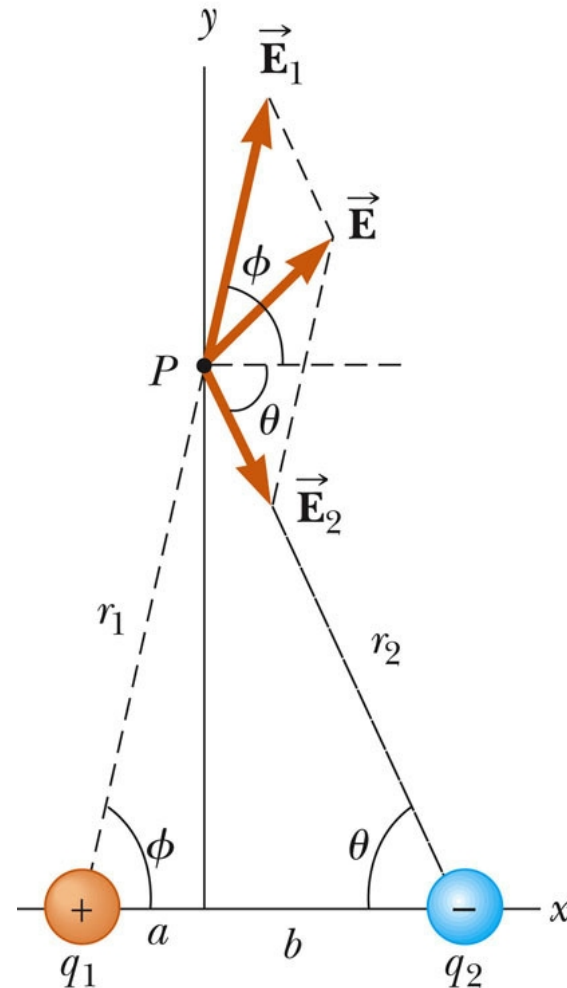
Or

$$\vec{E}_{net} = \sum_{i=1}^N \vec{E}_i = \vec{E}_1 + \vec{E}_2 + \dots + \vec{E}_N$$

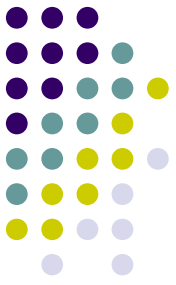


Superposition Example

- Find the electric field due to q_1 , $\vec{E}_1$
- Find the electric field due to q_2 , $\vec{E}_2$
- $\vec{E} = \vec{E}_1 + \vec{E}_2$
 - Remember, the fields add as vectors
 - The direction of the individual fields is the direction of the force on a positive test charge

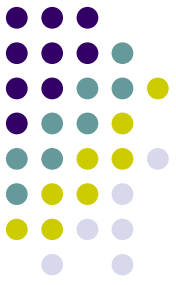


Electric Field – Continuous Charge Distribution

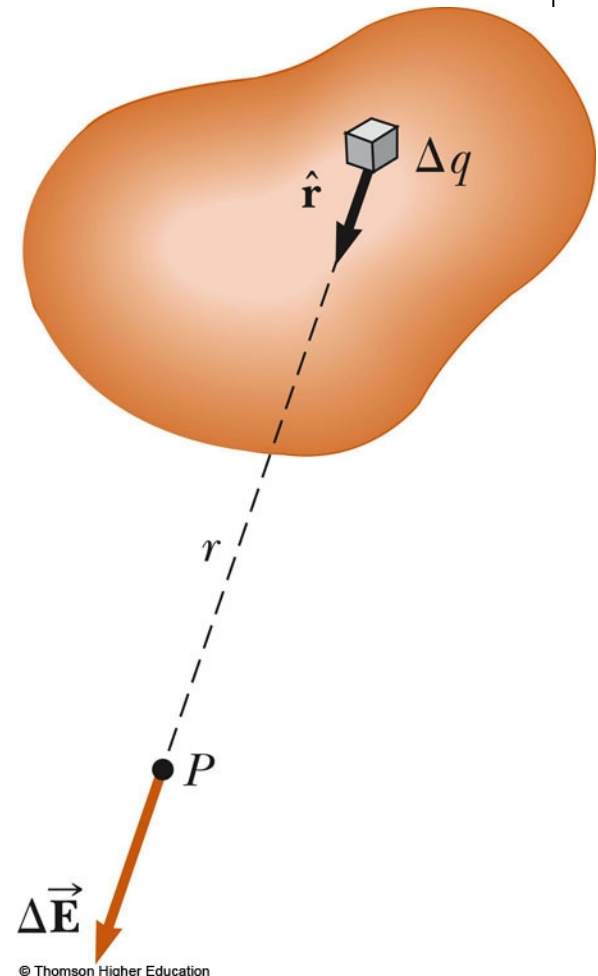


- The distances between charges in a group of charges may be much smaller than the distance between the group and a point of interest
- In this situation, the system of charges can be modeled as continuous
- The system of closely spaced charges is equivalent to a total charge that is continuously distributed along some line, over some surface, or throughout some volume

Electric Field – Continuous Charge Distribution, cont

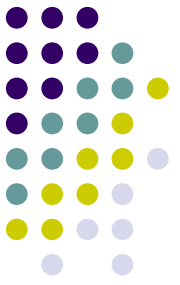


- Procedure:
 - Divide the charge distribution into small elements, each of which contains Δq
 - Calculate the electric field due to one of these elements at point P
 - Evaluate the total field by summing the contributions of all the charge elements



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Electric Field – Continuous Charge Distribution, equations

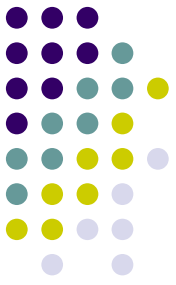


- For the individual charge elements

$$\Delta \vec{E} = k_e \frac{\Delta q}{r^2} \hat{r} \quad \text{so} \quad \vec{E}_{tot} = k_e \sum_i \frac{\Delta q_i}{r_i^2} \hat{r}_i$$

- Because the charge distribution is continuous, the sum becomes an integral over all charge:

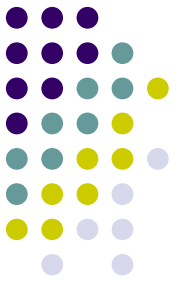
$$k_e \sum_i \frac{\Delta q_i}{r_i^2} \hat{r}_i \Rightarrow k_e \int \frac{dq}{r^2} \hat{r}$$



Charge Densities

- **Volume charge density:** when a charge is distributed evenly throughout a volume
 - $\rho \equiv Q / V$ with units C/m^3 ... so $dq = \rho dV$
- **Surface charge density:** when a charge is distributed evenly over a surface area
 - $\sigma \equiv Q / A$ with units C/m^2 ... so $dq = \sigma dA$
- **Linear charge density:** when a charge is distributed along a line (or arc)
 - $\lambda \equiv Q / \ell$ with units C/m ... so $dq = \lambda dl$

Amount of Charge in a Small Volume



- If the charge is nonuniformly distributed over a volume, surface, or line, the amount of charge, dq , is given by
 - For the volume: $dq = \rho \, dV$
 - For the surface: $dq = \sigma \, dA$
 - For the length element: $dq = \lambda \, d\ell$

Finding E due to continuous charge distributions



$$E = k_e \int \frac{dq}{r^2} \hat{r}$$

- 1) Substitute $dq = \lambda dx$, or σdA , ρdV , λdy , $\lambda r d\theta$, etc.
- 2) Substitute $r = x$ or y , if needed, to match dq .
- 3) Substitute $\hat{r} = \hat{i}$ or $\hat{j}$
- 4) Choose limits of integral to match dq and geometry
- 5) Use symmetry to simplify

Ex) Perhaps $E_x = 0$ or $E_y = 0$ so one integral is not needed

- 6) Place constants outside the integral
- 7) Integrate to find $E_x = \int dE_x$ and/or $E_y = \int dE_y$
- 8) Combine into vector expression

$$\vec{E} = E_x \hat{i} + E_y \hat{j} \quad \text{or} \quad \vec{E} = E_r \hat{r}$$

PHYS 2321: Physics 2

Week 3 -- Electric Fields & Charge Distributions



Day 3 outline

1) Homework for next Wednesday

a) Next: Ch. 22 P. 1,2,5,6,9,10,13,17,19,20,35

MisCQs 1-9 (odd)

2) Today: Electric Fields

a) Calculate E due to disk charge and infinite plane charge.

b) Field lines

c) Motion of charges in E-fields (P. 57)

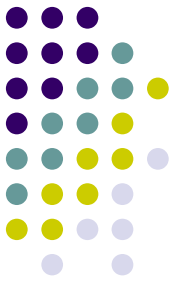
d) Electric Flux $\Phi = \int \vec{E} \cdot d\vec{A}$

e) Gauss's Law $\oint \vec{E} \cdot d\vec{A} = \frac{Q_{enc}}{\epsilon_0}$

Note: Physics Tutor on 2nd floor Library on Thurs 7-9 pm.

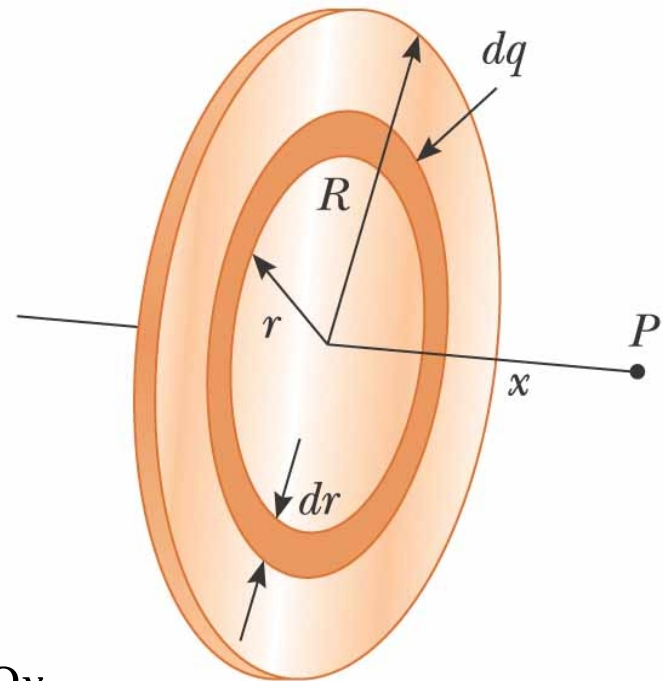
Canvas published for use of grade book

Next quiz: Monday (on E-fields).



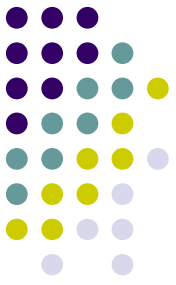
Example – Charged Disk

- The ring has a radius R and a uniform charge density σ
- Choose dq as a ring of radius r
- The ring has a surface area $2\pi r dr$
- So $dq = 2\pi r \sigma dr$
- Build on “Ring of charge” Example



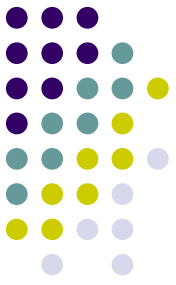
$$E = \frac{1}{4\pi\epsilon_0} \frac{Qx}{(x^2 + a^2)^{3/2}}$$

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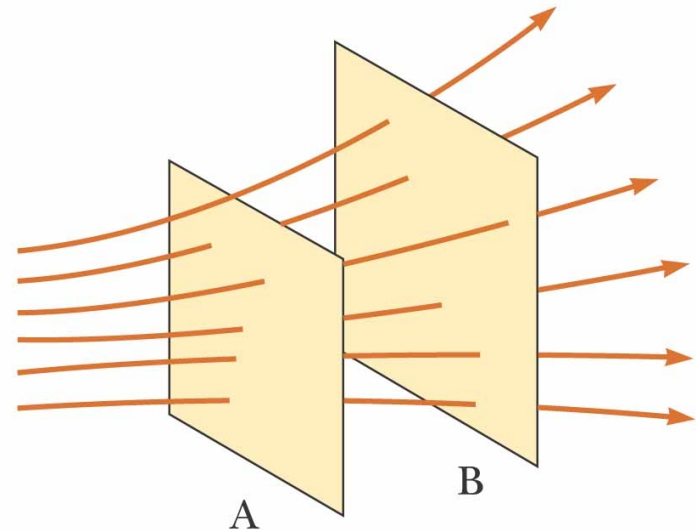
Electric Field Lines

- Field lines give us a means of representing the electric field pictorially
- The electric field vector $\mathbf{E}$ is tangent to the electric field line at each point
 - The line has a direction that is the same as that of the electric field vector
- The number of lines per unit area through a surface perpendicular to the lines is proportional to the magnitude of the electric field in that region



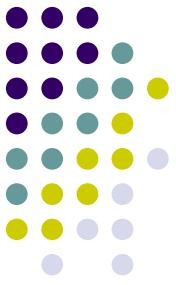
Electric Field Lines, General

- The density of lines through surface A is greater than through surface B
- The magnitude of the electric field is greater on surface A than B
- The lines at different locations point in different directions
 - This indicates the field is nonuniform

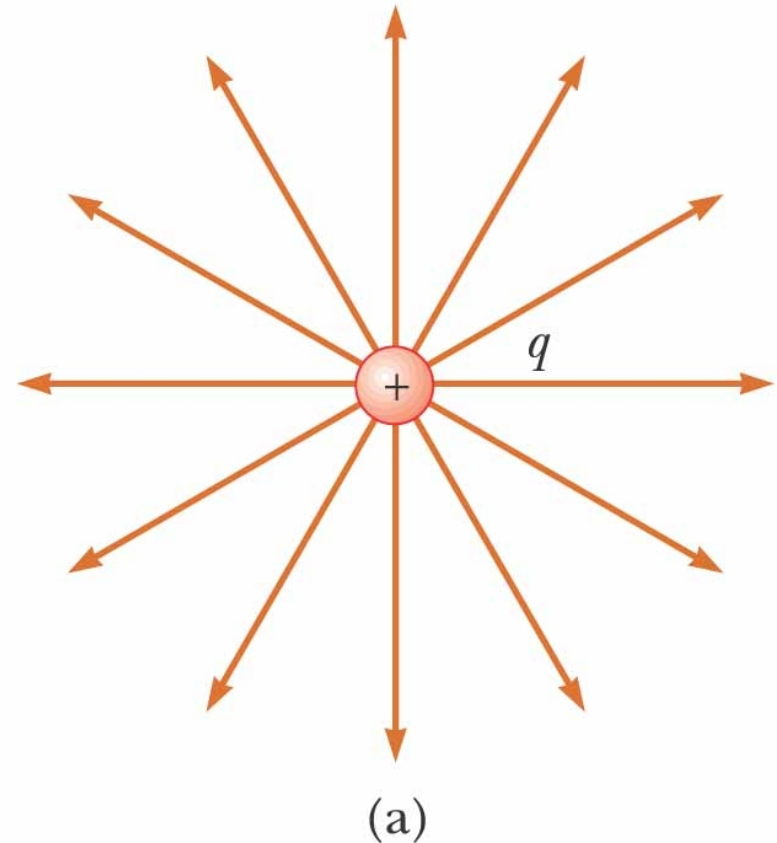


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Electric Field Lines, Positive Point Charge

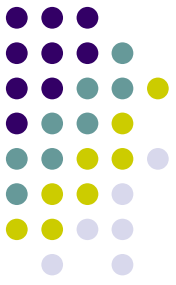


- The field lines radiate outward in all directions
 - In three dimensions, the distribution is spherical
- The lines are directed away from the source charge
 - A positive test charge would be repelled away from the positive source charge

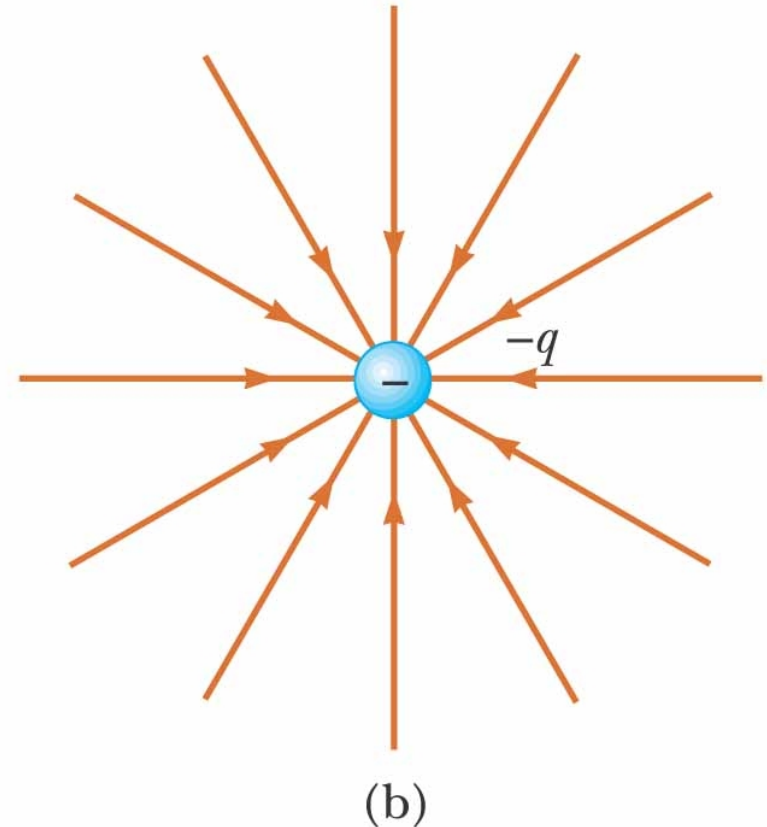


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Electric Field Lines, Negative Point Charge

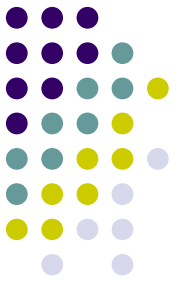


- The field lines radiate inward in all directions
- The lines are directed toward the source charge
 - A positive test charge would be attracted toward the negative source charge

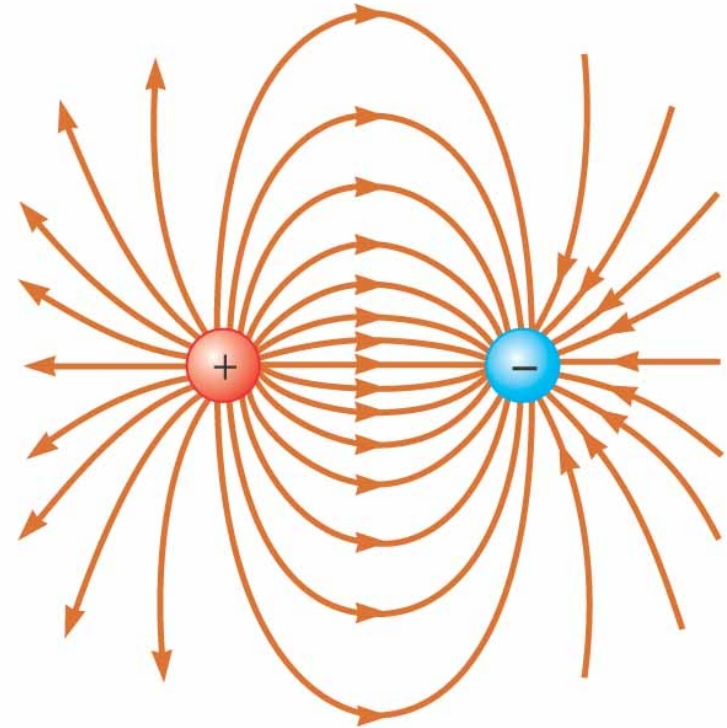


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Electric Field Lines – Dipole

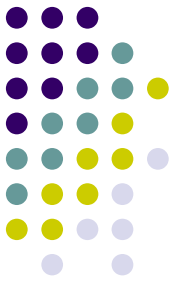


- The charges are equal and opposite
- The number of field lines leaving the positive charge equals the number of lines terminating on the negative charge

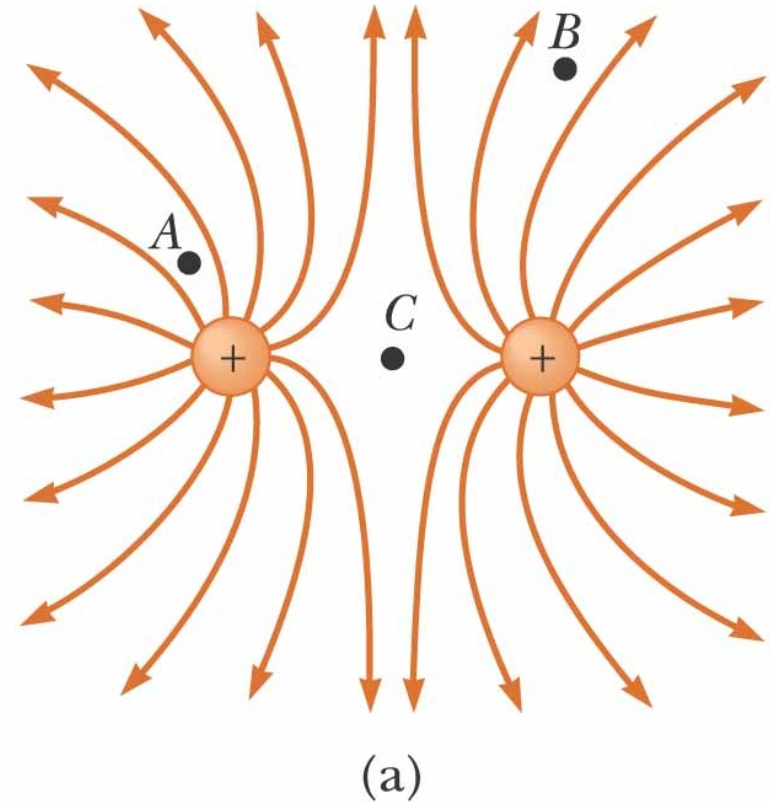


(a)

Electric Field Lines – Like Charges

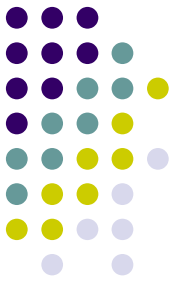


- The charges are equal and positive
- The same number of lines leave each charge since they are equal in magnitude
- At a great distance, the field is approximately equal to that of a single charge of $2q$

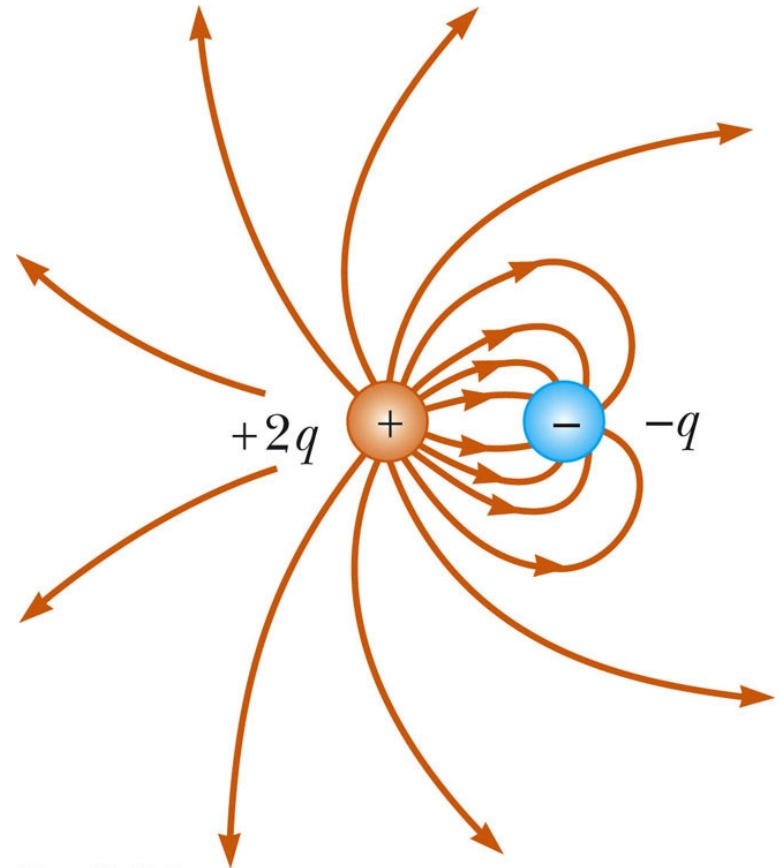


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Electric Field Lines, Unequal Charges

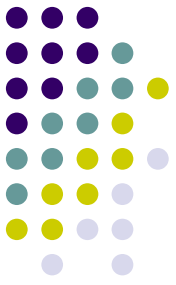


- The positive charge is twice the magnitude of the negative charge
- Two lines leave the positive charge for each line that terminates on the negative charge
- At a great distance, the field would be approximately the same as that due to a single charge of $+q$
- Use the active figure to vary the charges and positions and observe the resulting electric field



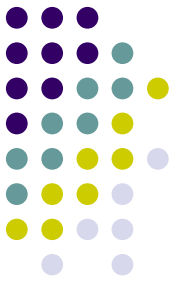
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Electric Field Lines – Rules for Drawing*



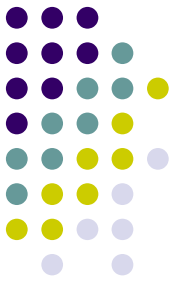
- The lines must begin on a positive charge and terminate on a negative charge
 - In the case of an excess of one type of charge, some lines will begin or end infinitely far away
- The number of lines drawn leaving a positive charge or approaching a negative charge is proportional to the magnitude of the charge
- No two field lines can cross
- Remember field lines are **not** material objects, they are a pictorial representation used to qualitatively describe the electric field

* See my 9 rules on my YouTube video.



Motion of Charged Particles

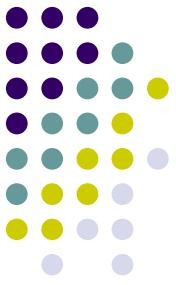
- When a charged particle is placed in an electric field, it experiences an electrical force
- If this is the only force on the particle, it must be the net force
- The net force will cause the particle to accelerate according to Newton's second law



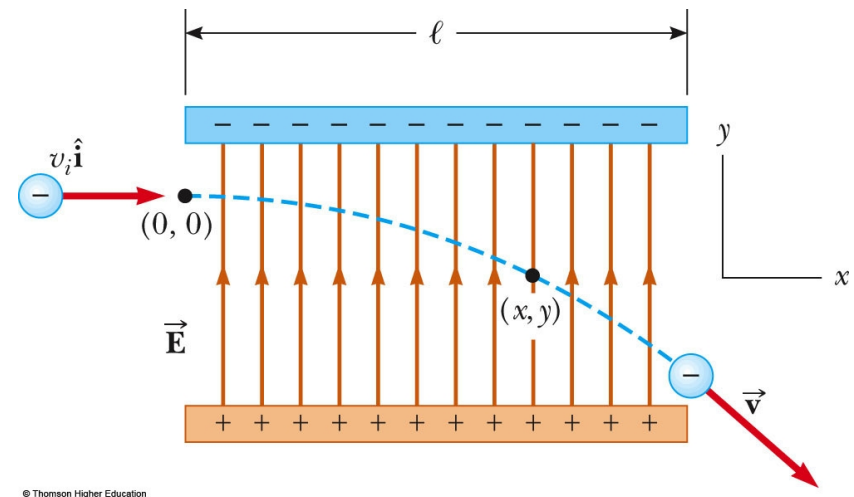
Motion of Particles, cont

- When only Coulomb force exists, $\mathbf{F} = q\mathbf{E} = m\mathbf{a}$.
- If $\mathbf{E}$ is uniform, then the acceleration is constant.
- If the particle has a positive charge, its acceleration is in the direction of the field.
- If the particle has a negative charge, its acceleration is in the direction opposite the electric field.
- If the acceleration is constant, the kinematic equations for uniform acceleration can be used.

Electron in a Uniform Field, Example



- The electron is projected horizontally into a uniform electric field
- The electron undergoes a downward acceleration
 - It is negative, so the acceleration is opposite the direction of the field
- Its motion is parabolic while between the plates



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Use the active figure to vary the field and the characteristics of the particle.